Fiscal Year:	FY 2015	**Task Last Updated:**	FY 01/26/2016
PI Name:	Elefteriou, Florent Ph.D.		
Project Title:	Vestibular Stimulation: A Countermeasure For Bone Loss During Long-Term Space Mission?		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	37232-0575	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	07/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Besnard, Stephane (INSERM, University of Caen Basse-Normandie (FRANCE)) Denise, Pierre (Université de Caen Basse-Normandie) Makishima, Tomoko (University of Texas, Galveston) Perrien, Daniel (Vanderbilt University)		
Grant/Contract No.:	NNX12AL35G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Background: We have demonstrated that loss of vestibular information following bilateral lesions in rats and mice decreases lower limb bone density, and that this effect may be mediated by activation of the sympathetic nervous system (SNS). Based on the knowledge that reduced vestibular information, increased sympathetic activation, and bone loss are common hallmarks of long-term space missions, we hypothesized that space-induced bone loss has a vestibular and sympathetic origin, and that vestibular stimulation might increase bone density. The main objective of this proposal is to evaluate the effect of vestibular stimulation on bone mass, with the goal of designing a new therapeutic approach to prevent bone loss in space. The project is divided into two specific aims. The first one will define the short and long-term effects of vestibular hair cell dysfunction on the vegetative system and on bone mass in mice, in order to determine whether this effect is compensated with time. Catecholamine levels, brown adipose tissue Ucp1/Gapdh expression, 3D-microcomputed tomography, and quantitative static and dynamic bone histomorphometry will be used. The second specific aim will characterize the effects of vestibular stimulation on the vegetative system and bone. Different centrifuge protocols (continuous and discontinuous, 1.5G/2G/2.5G) and galvanic vestibular stimulation will be used. Centrifugations will be performed in Dr. Makishma (University of Texas Medical Branch (UTMB), USA) and in Dr. Besnard and Prof. Denise (University of Caen, France) laboratories. Galvanic stimulation will be performed in Dr. Elefteriou's laboratory, in collaboration with Dr. Wood (Houston/UTMB, USA). All the bone analyses will be performed at Vanderbilt University (Dr. Elefteriou, Vanderbilt University). The part of the study performed in France is supported by the University of Caen Basse-Normandie and by the Centre national d'études spatiales (CNES). The ultimate goal of this multi-disciplinary effort is to determine whether vestibular stimulation could have beneficial effects on bone mineral density and sympathetic function during extended space missions. These findings may have application on Earth as well for the treatment or prevention of low bone mass conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research seeks to understand if and how structural and functional alterations in the vestibular system lead to changes in bone remodeling. This work is expected to impact our understanding of microgravity-induced bone loss, which is relevant to extended travel duration of future space missions, but also to fracture risk in patients with vestibular disorders on Earth.
Task Progress:	Aim1. To define the short and long-term effects of vestibular loss on the vegetative system and on bone mass in mice. This Aim is completed and results were reported in the Journal of Bone and Mineral Research in December 2014 (final published version June 2015-- http://dx.doi.org/). Aim 2. To characterize the effects of vestibular stimulation on the vegetative system and bone. a) Centrifugations. Four to five month-old het/+ (Control) and het/het mice (devoid of otoconia from birth) submitted to continuous centrifugation at 2G for 4 weeks in Dr. Besnard's laboratory (France) did not experience any significant bone changes at the distal femur, as assessed by 3D computed microtomography (12 um resolution). Bone volume/tissue volume and tissue mineral density were indeed not significantly different between groups (n=10 per group). After a setback of several months to get the centrifuge operational in Dr. T. Makishima's laboratory, 4 month-old female WT mice have been submitted to intermittent centrifugation (2h/day) at 2G for 4 weeks. Here again, no differences were detected in terms of bone volume/tissue volume and tissue mineral density between control and treated group, as assessed by 3D computed microtomography (n=10 per group). Because of the negative nature of these results, this work has not been submitted for publication yet. b) Galvanic vestibular stimulation. The Hindlimb Suspension (HS) model has been set up successfully in collaboration with Dr. D. Perrien at Vanderbilt. Following our discussions with Dr. Wood and Dr. Makishima in Galveston, we have determined the conditions of GVS stimulation that can induce nystagmus, using an eye tracker recording system. Then, WT mice have been tested for their response to daily GVS stimulations, in absence or presence of hindlimb suspension (HS). HS, as expected, triggered a significant bone loss in distal femurs of WT mice, as assessed by 3D computed microtomography. However, daily GVS stimulations for 21 days did not significantly alter bone mass. Daily GVS stimulations under HS conditions did not affect bone mass either (no protection from bone loss nor further bone loss).
Bibliography Type:	Description: (Last Updated: 08/16/2018)
Abstracts for Journals and Proceedings	Zuniga MG, Ladner T, Vignaux G, Hoffman H, Elefteriou F, Rivas A. "Lower Bone Mineral Density in Weight-bearing Bones amongst Individuals with Generalized Vestibular Weakness." 38th Annual MidWinter Meeting of ARO (Association for Research in Otolaryngology), Baltimore, MD, February 21-25, 2015. 38th Annual MidWinter Meeting of ARO (Association for Research in Otolaryngology), Baltimore, MD, February 21-25, 2015. , Feb-2015
Abstracts for Journals and Proceedings	Vignaux G, Ndong JD, Perrien DS, Elefteriou F. "Impact of vestibular alterations on bone remodeling." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Articles in Peer-reviewed Journals	Elefteriou F. "Impact of the autonomic nervous system on the skeleton." Physiol Rev. 2018 Jul 1;98(3):1083-112. Review. Epub 2018 May 2. https://doi.org/10.1152/physrev.00014.2017 ; PubMed PMID: 29717928; PubMed Central PMCID: PMC6088147 , Jul-2018
Articles in Peer-reviewed Journals	Vignaux G, Ndong JD, Perrien DS, Elefteriou F. "Inner ear vestibular signals regulate bone remodeling via the sympathetic nervous system." J Bone Miner Res. 2015 Jun;30(6):1103-11. [Epub 2014 Dec 10] http://dx.doi.org/10.1002/jbmr.2426 ; PubMed PMID: 25491117 , Jun-2015

Articles in Peer-reviewed Journals	Vignaux G, Besnard S, Denise P, Elefteriou F. "The vestibular system: a newly identified regulator of bone homeostasis acting through the sympathetic nervous system." Current Osteoporosis Reports. 2015 Aug;13(4):198-205. [Epub 2015 May 28] http://dx.doi.org/10.1007/s11914-015-0271-2 ; PubMed PMID: 26017583 , Aug-2015